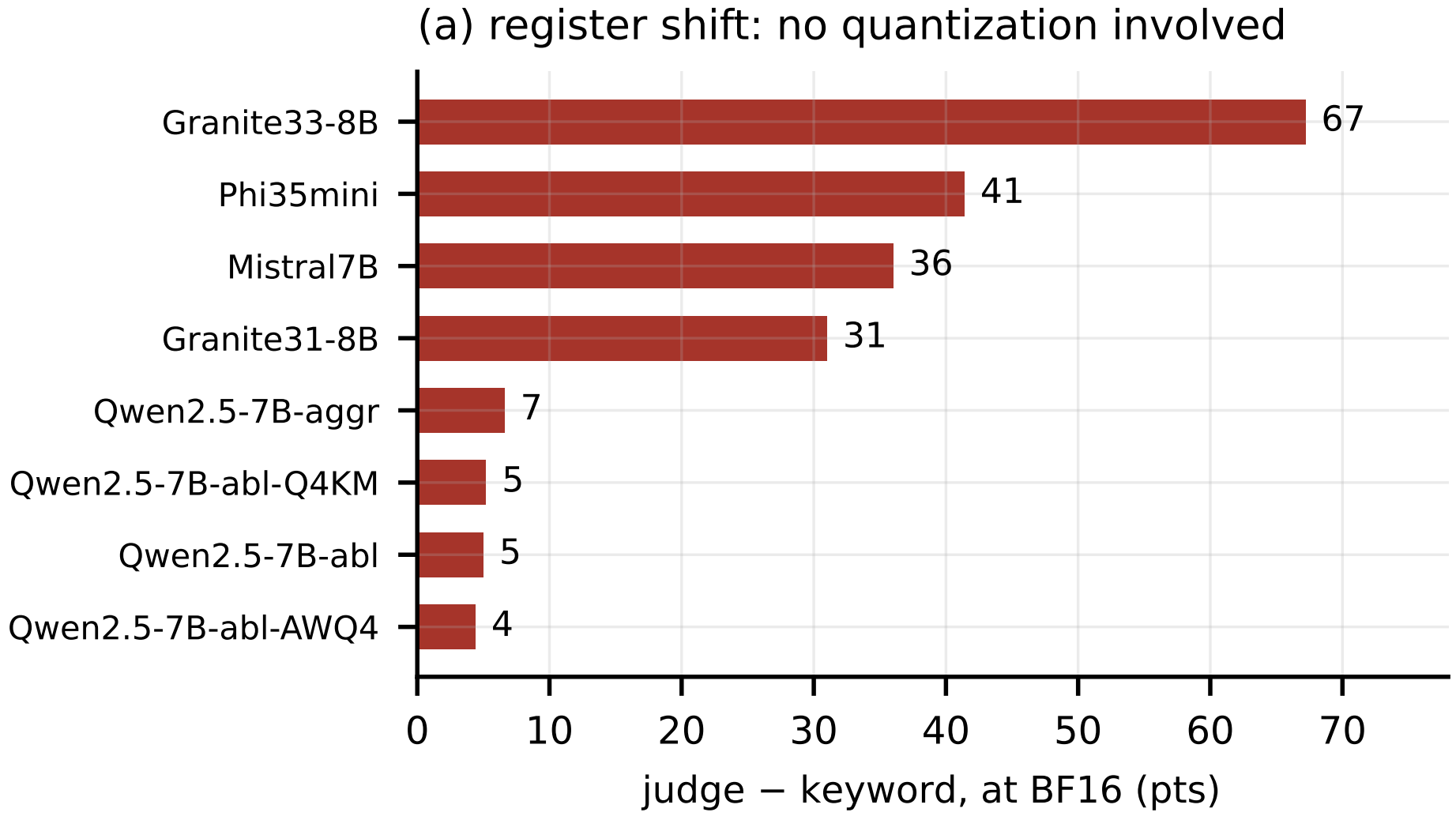
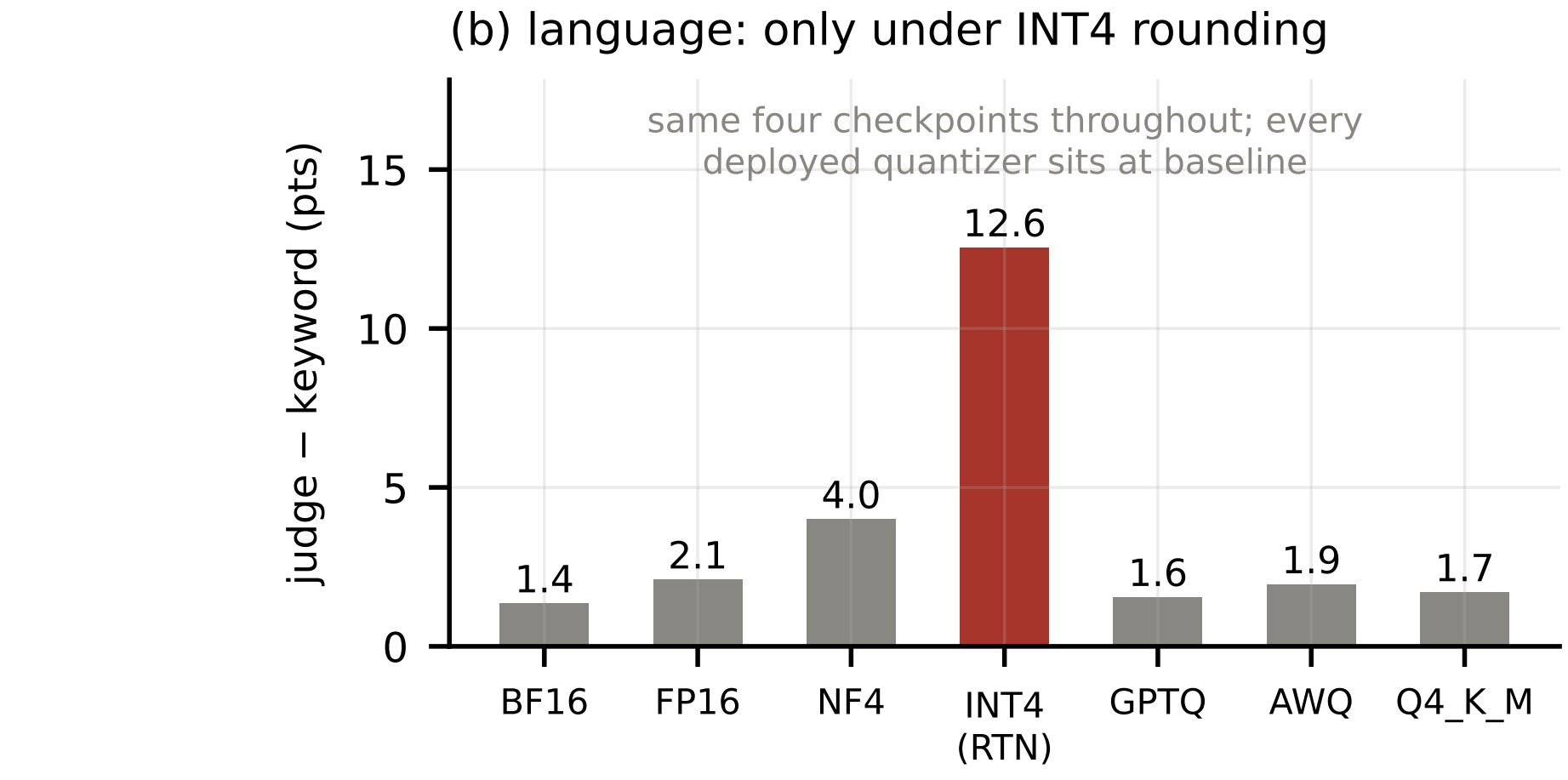

(a) register shift: no quantization involved
Granite33-8B
Phi35mini
Mistral7B
Granite31-8B
Qwen2.5-7B-aggr
Qwen2.5-7B-abl-Q4KM
Qwen2.5-7B-abl
Qwen2.5-7B-abl-AWQ4
67
41
36
31
7
5
5
4
0
10
20
30
40
50
60
70
judge − keyword, at BF16 (pts)


(b) language: only under INT4 rounding
same four checkpoints throughout; every deployed quantizer sits at baseline
judge − keyword (pts)
0
5
10
15
1.4
2.1
4.0
12.6
1.6
1.9
1.7
BF16
FP16
NF4
INT4 (RTN)
GPTQ
AWQ
Q4_K_M